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COURSE: Design and Analysis of Algorithms

CODE: CSA0619

REPORT

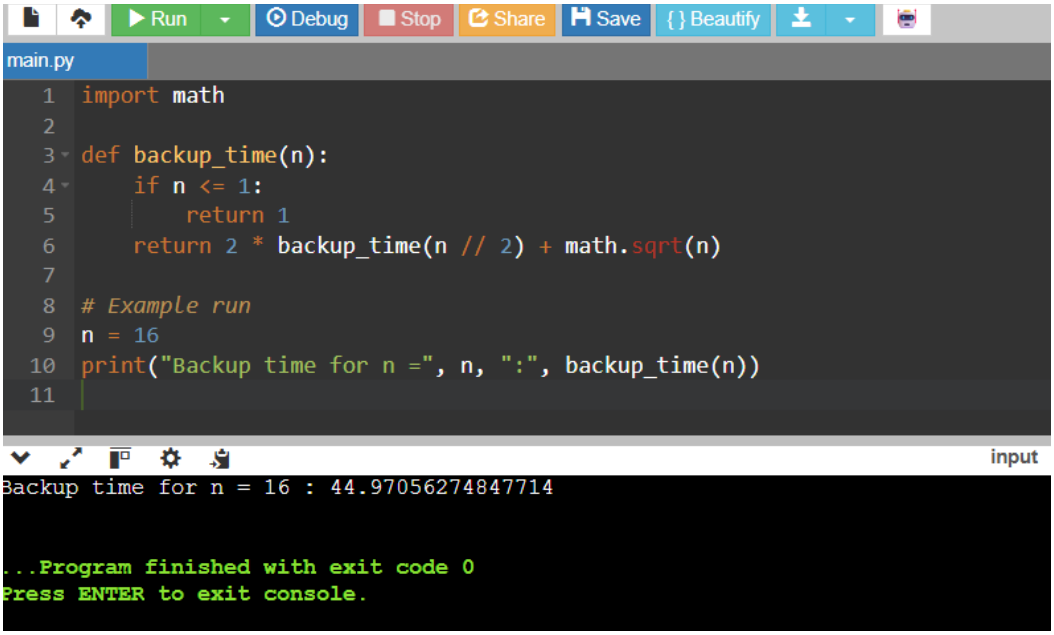
Q9. Application: Data Backup Optimization System

A backup system uses an algorithm with recurrence T(n) = 2T(n/2) + √n. Apply the Master Theorem to determine the time complexity. Compare f(n) with n^log_b a and identify the correct case. Analyze performance for large datasets. Interpret efficiency.

1. **Source Code (Python Example)**

```
import math
def backup_time(n):
    if n <= 1:
        return 1
    return 2 * backup_time(n // 2) + math.sqrt(n)
# Example run
n = 16
print("Backup time for n =", n, ":", backup_time(n))
```

2.OUTPUT



We are given the recurrence:

$$T(n) = 2T\left(\frac{n}{2}\right) + \sqrt{n}$$

- Parameters:
  - $a = 2$(number of subproblems)
  - $b = 2$(factor by which problem size is reduced)
  - $f(n) = \sqrt{n}$

Now compute:

$$n^{\log_b a} = n^{\log_2 2} = n^1 = n$$

So we compare $f(n) = \sqrt{n}$with n.

- Case 1: If $f(n) = O(n^c)$where $c < \log_b a$, then $T(n) = \Theta\left(n^{\log_b a}\right)$.
- Case 2: If $f(n) = \Theta\left(n^{\log_b a}\right)$, then $T(n) = \Theta\left(n^{\log_b a} \log n\right)$.
- Case 3: If $f(n) = \Omega(n^c)$where $c > \log_b a$, then $T(n) = \Theta\left(f(n)\right)$.

Here:

- $f(n) = \sqrt{n} = n^{1/2}$

- $\log_b a = 1$
  So $f(n) = n^{1/2}$, which is **polynomially smaller** than n^1.
  ☞ This matches **Case 1**.

T(n)=Θ(n)

ANALYSIS:

- For **large datasets**, the algorithm runs in **linear time**.
- The additional cost of $\sqrt{n}$per level is negligible compared to the dominant n.
- This means the backup system scales efficiently, even with millions of files.

Q25. Application: Fraud Detection System

A fraud detection algorithm follows the recurrence T(n)=2T(n/2)+n 2 logn. Apply the Master Theorem to determine the time complexity. Compare f(n) with n log b a and justify the correct case. Analyze performance for large transaction datasets. Interpret efficiency clearly.

1.Source Code (Python):

```
import math
def fraud_detection_time(n):
   # Base case
   if n <= 1:
```

```
        return 1

    # Recurrence relation: T(n) = 2T(n/2) + n^2 log n
    return 2 * fraud_detection_time(n // 2) + (n**2) *
math.log(n)

# Example run
for n in [2, 4, 8, 16]:
    print(f"T({n}) = {fraud_detection_time(n):.2f}")
```

2. OUTPUT:

We are given:

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2 \log n$$

- $a = 2$(number of subproblems)
- $b = 2$(factor by which problem size is reduced)
- $f(n) = n^2 \log n$

Compare with $n^{\log_b a}$

$$n^{\log_b a} = n^{\log_2 2} = n^1 = n$$

So we compare $f(n) = n^2 \log n$with n.

Clearly:

$$n^2 \log n \gg n$$

That means $f(n)$grows **polynomially faster** than $n^{\log_b a}$.

- **Case 1:** $f(n)$smaller → not applicable.
- **Case 2:** $f(n)$same order → not applicable.
- **Case 3:** $f(n)$larger → applicable.

👉 This matches **Case 3**.

T(n)=Θ(n2logn)

ANALYSIS:

- For **large transaction datasets**, the algorithm runs in **super-quadratic time**.
- The $n^2 \log n$term dominates, meaning runtime grows faster than quadratic but slower than cubic.
- As dataset size doubles, runtime increases more than fourfold due to the logarithmic multiplier.